BUSINESS DECISION ENGINE

ABSTRACT

A tool that understands your marketing spend and returns based on channels and gives accurate analytics.

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Introduction

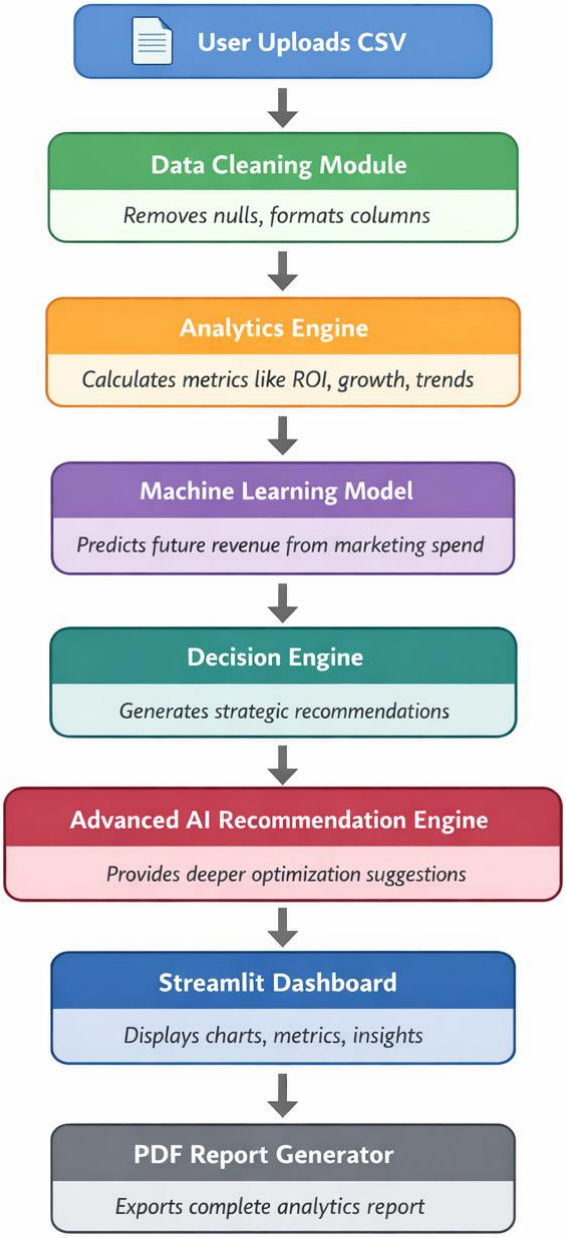
The Business Decision Engine is a data analytics and AI-powered decision support tool designed to analyse marketing performance data and generate actionable business insights.

The tool allows users to upload marketing datasets (CSV format), automatically clean and process the data, analyse key performance metrics, run machine learning predictions, and generate strategic recommendations. The results are displayed in an interactive dashboard and can also be exported as a structured PDF report.

This system combines data analytics, machine learning, and decision automation to help users understand marketing performance and optimize budget allocation across channels.

System Flow

The overall workflow of the tool is shown below.



How the Tool Works

The system operates through the following steps:

Step 1: Dataset Upload

The user uploads a marketing dataset through the Streamlit dashboard. The dataset typically contains fields such as:

- Date
- Marketing channel
- Marketing spend
- Revenue generated

Note: As per the code its "date", "marketing_spend", "revenue" and "channel".

Step 2: Data Cleaning

The dataset is sent to the backend where it is cleaned and formatted to remove missing or inconsistent data.

Step 3: Analytics Processing

The analytics engine processes the cleaned dataset to calculate important business metrics such as:

- Total marketing spend
- Total revenue
- Return on Investment (ROI)
- Growth trends
- Channel performance

Step 4: Machine Learning Prediction

A machine learning regression model analyses historical spend and revenue patterns to estimate future revenue outcomes.

Step 5: Decision Engine Processing

The decision engine evaluates:

- Channel performance
- ROI trends
- ML predictions

Using these results, it generates strategic business recommendations.

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Step 6: AI Recommendations

An advanced AI module further analyses the insights and produces high-level recommendations such as:

- Increasing spend on high-performing channels
- Reducing spend on underperforming campaigns
- Optimizing marketing allocation strategies

Step 7: Dashboard Visualization

The Streamlit dashboard presents the results through:

- KPI metrics
- Performance charts
- Channel comparison graphs
- Budget allocation visualizations
- AI recommendations

Step 8: Report Generation

Users can download a complete analytics report in PDF format that includes:

- Dataset summary
- Key metrics
- Marketing insights
- ML predictions
- Decision engine outputs
- AI recommendations

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Usage:

Follow the step-by-step guidelines to set this tool efficiently:

1. git clone https://github.com/J0hns0nJ0y/Business-Decision-Engine.git
2. cd Business-Decision-Engine
3. venv\Scripts\activate
4. pip install -r requirements.txt

Your tool is all set for execution

Run backend server:

Once you are inside the repository folder

1. cd backend
2. uvicorn main:app --host 0.0.0.0 --port 8000 --reload
3. go to any browser and type "127.0.0.1:8000/docs"
4. Click on "**POST** /upload" drop down.
5. Click on "Try It Out" button.
6. Choose a .csv file and upload it. Then Execute it.

You'll get a Json file of analysis.

Run frontend server:

Once your backend server is running properly, open another terminal and follow these steps

1. cd Business-Decision-Engine
2. venv\Scripts\activate
3. cd frontend
4. streamlit run dashboard.py --server.address 0.0.0.0 –server.port 8501
5. go to any browser and type "127.0.0.1:8501" OR it shall openly automatically.
6. Upload your CSV file

Suppose you want to download a report, a button is available at the bottom of UI dashboard.

Ensure the backend server is running on Terminal 1 & front-end server is running on Terminal 2.

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Limits:

- The tool is limited to certain columns in excel only, User will have to rename columns as "date", "marketing_spend", "revenue" and "channel". Required columns must be in lowercase. This limitation isn't permanent as developers can change the code in backend files.

- The tool doesn't predict and refill columns that have error. Instead, it removes those error rows and does analysis based on available rows.

Analyst Usage Method:

- ✓ To understand numerical performance, analyst may prefer viewing from the backend server JSON file generated. As this file produces numerical calculations output.
- ✓ To understand the performance visually, the frontend dashboard will be used. A good analyst uses both methods to understand the company performance.

Contribute:

If you are willing to contribute your ideas into this project feel free to raise a issue or pull requests at

https://github.com/J0hns0nJ0y/Business-Decision-Engine.git

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